

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

MA-111-CBCS

B.E. I & II Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Mathematics - II

Time : Three Hours

Maximum Marks: 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Solve $\frac{dx}{dt} + 4x + 3y = t$
 $\frac{dy}{dt} + 2x + 5y = e^t$

2. a) Find the rank of matrix A by Echelon form where

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix} \text{ and also find its nullity of A.}$$

- b) Solve the following equations

$$x + 2y + 3z = 0$$

$$2x + 3y + z = 0$$

$$4x + 5y + 4z = 0$$

$$x + 2y - 2z = 0$$

MA-111-CBCS

PTO

[2]

3. a) Solve $(1+y^2) dx = (\tan^{-1}y - x) dy$
b) Solve $(D^2 - 3D + 2)y = e^{3x}$ where $D = \frac{d}{dx}$
4. a) Solve $xzp + yzq = xy$
b) Solve by Charpit method $z = px + qy + p^2 + q^2$
5. a) Solve $\frac{dy}{dx} + y = 1$
b) Solve $(D^3 - 3D^2 + 4)y = 0$
6. Solve $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x$
7. a) Solve $(D^2 - 2DD' + D'^2)z = e^{n+2y} + x^3$
b) Solve $\frac{\partial u}{\partial x} = \frac{2\partial u}{\partial t} + u$ by the method of separation of variables.

OR

Verify Cayley-Hamilton theorem for the following matrix and hence find A^{-1}

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

8. Solve $x^2p + y^2q = (x + y)z$

MA-111-CBCS